A nonconvex incenter locus with nested circular caustics

A counterexample to MathDB 351604

Principia Math in Progress

12 September 2026

Task 1. Resolve the bic-III convexity conjecture

Conjecture 2 of [1, p. 23] asserts that the incenter locus of a bic-III triangle family is a convex curve. Its four circles must be distinct. We use *convex curve* in the usual boundary-of-a-convex-set sense, rather than requiring that the curve itself be a convex set.

Theorem 1. *There exists a bic-III family with all three circular caustics strictly inside its circumcircle, and with no degenerate triangles, whose incenter locus is not a convex curve.*

Proof. Put $h = 2/5$, $r = 7/12$, and $A = 1 + h^2 - r^2 = 2951/3600$. Consider the pencil

$$C_\lambda : \quad x^2 + y^2 - 2\lambda h x - 1 + \lambda A = 0.$$

Its center is $(\lambda h, 0)$ and its squared radius is $1 - \lambda A + \lambda^2 h^2$. The circumcircle is C_0, the unit circle. Use C_1 and $C_{7/4}$ as the first two caustics. Their centers and radii are

$$O_1 = (2/5, 0), \quad r_1 = 7/12, \qquad O_2 = (7/10, 0), \quad r_2 = \sqrt{799}/120.$$

They are strictly nested: $2/5 + 7/12 < 1$ and $3/10 + \sqrt{799}/120 < 7/12$. Their centers are distinct.

Here is a completely specified family, including the tangent choices. Set

$$P(t) = \left(\frac{1 - t^2}{1 + t^2}, \frac{2t}{1 + t^2} \right), \qquad J(x, y) = (-y, x), \qquad t \in \mathbb{R} \cup \{\infty\}.$$

For a circle with center O, radius ρ, and sign $s \in \{-1, 1\}$, define

$$V = O - P, \qquad D = \langle V, V \rangle, \qquad W = \frac{\sqrt{D - \rho^2}\, V + s\rho J V}{D}, \tag{1}$$

$$T(O, \rho, s; t) = P - 2\langle P, W \rangle W. \tag{2}$$

Take $Q(t) = T(O_1, r_1, -1; t)$ and $Z(t) = T(O_2, r_2, 1; t)$. The vector W has length one, and the distance from O to the line $P + \mathbb{R}W$ is ρ. Equation (2) gives its second intersection with the unit circle. Thus PQ and PZ are the specified tangent chords. The square roots are positive everywhere because the caustics lie strictly inside the unit circle. Also P, Q, Z are pairwise distinct: $Q = Z$ would give a common tangent to the two strictly nested caustics. Three distinct points on a circle are noncollinear. Consequently the family and its incenter are smooth, including through the parameter value at infinity.

Poncelet's Main Lemma, [1, Lemma 1], gives a fixed in-pencil envelope for QZ. We verify explicitly that it is a third proper inner circle, distinct from the first two. For either $t = 0$ or $t = 1$, let

$$n = J(Z - Q), \qquad e = \langle n, Q \rangle.$$

The line QZ is tangent to C_λ precisely when

$$g_t(\lambda) := (e - \lambda h n_x)^2 - (1 - \lambda A + \lambda^2 h^2)\langle n, n \rangle = 0. \tag{3}$$

Exact rational interval evaluation gives the following isolating intervals for the two roots; all endpoints are divided by 10^9:

t	smaller root	larger root
0	$(1957015893, 1957015894)$	$(11272411549, 11272411550)$
1	$(1957015893, 1957015894)$	$(3366427888, 3366427889)$

The coefficients of λ^2 are nonzero. Since the larger-root intervals are disjoint, the fixed envelope supplied by the lemma must be C_{λ_*}, where λ_* is the smaller root of g_0. In particular $1.95 < \lambda_* < 1.96$. Throughout that interval the squared radius is between 0.001744444 and 0.016197667. Writing its radius as r_*, the same interval evaluation gives

$$h\lambda_* + r_* < 0.912 < 1, \qquad h\lambda_* - 7/10 + r_* < 0.212 < r_2.$$

Thus the third caustic is strictly inside $C_{7/4}$. Its parameter differs from $0, 1, 7/4$, so all four circles are distinct.

The incenter is, using ordinary positive side lengths,

$$I(t) = \frac{aP + bQ + cZ}{a + b + c}, \quad a = |Q - Z|, \quad b = |P - Z|, \quad c = |P - Q|. \tag{4}$$

For vectors u, v, write $[u, v] = u_x v_y - u_y v_x$. The following exact enclosures suffice for the counterexample:

quantity	enclosure
$I'_x(-1/3)$	$(206784605, 206784606)/10^9$
$[I'(-1/3),\ I(0) - I(-1/3)]$	$(14992242, 14992243)/10^9$
$[I'(-1/3),\ I(-1/2) - I(-1/3)]$	$(-144056, -144055)/10^9$

The first row shows that the tangent is nonzero. The next two rows put points of the same incenter locus strictly on opposite sides of that tangent line. If this locus were the boundary of a convex set, a supporting line at $I(-1/3)$ would have to be its tangent: apply the support inequality to $I(-1/3 + \varepsilon)$ for both signs of ε and differentiate at zero. All boundary points would then lie in one closed half-plane, contradicting the two signs.

For completeness, these are rigorous integer interval computations, not floating-point sign tests. The accompanying `verify.py` evaluates (1)–(4) with outward rounding to denominator $B = 10^{60}$ after every operation. An interval is stored as integer endpoints $[l, u]/B$. Multiplication rounds the minimum and maximum endpoint products outwards after division by B; division does the same with the four endpoint quotients, rejecting a denominator containing zero. Positive square roots are enclosed by

$$\left[\frac{\lfloor \sqrt{lB} \rfloor}{B}, \frac{\lceil \sqrt{uB} \rceil}{B} \right].$$

The integer square-root bounds are checked by squaring. Derivatives are evaluated with pairs (f, f') and the product, quotient and square-root differentiation rules, starting with $(t, 1)$. The quadratic formula applied to (3) gives the displayed root intervals. These operations, using only integer arithmetic and rational quotients, reproduce every strict bound above. □

The related paper [2] discusses cusps caused by exterior caustics and degenerate triangles. The present counterexample uses strictly nested proper circles and a smooth nondegenerate family; it does not rely on those phenomena.

Task 2. Determine the convex parameter region

Characterize the pairs of nested caustics in a circular pencil for which a selected bic-III incenter locus is convex, taking account of all four choices of tangent chords.

Task 3. Describe the onset of nonconvexity

Find exact parameter conditions for the first vanishing of the signed curvature, and determine which bifurcations preserve regularity and simplicity of the incenter locus.

Task 4. Determine the algebraic locus

Compute a minimal implicit equation and its degree for the incenter locus of this example, separating the real incenter branch from the excenter branches introduced by squaring.

Task 5. Compare the four triangle families

Determine how convexity, inflections, and winding numbers change under the four tangent choices associated with fixed first and second caustics.

References

[1] R. Garcia, B. Odehnal and D. Reznik, *Loci of Poncelet triangles in the general closure case*, J. Geom. **113** (2022), article 17, 30 pp., doi:10.1007/s00022-022-00629-3. Conjecture 2, p. 23; arXiv:2108.05430v2, p. 16.

[2] R. Alves Garcia, B. Odehnal and D. Reznik, *Poncelet porisms in hyperbolic pencils of circles*, J. Geom. Graphics **25** (2021), 205–225.